

TECHNOLOGY STACK

Date	29-09-2026
Team ID	SWTID-2026-6684
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	5 Marks

Define Your Technology Stack

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	React.js / Next.js Tailwind CSS HTML5 Canvas	• React/Next.js: Enables responsive, component-based UI rendering for fast visual comic creation and dynamic panel updates. • Tailwind CSS: Streamlines visual layout customization, speech bubble styling, and dark/light UI mode handling. • HTML5 Canvas: Allows interactive drag-and-drop elements, custom panel slicing, and visual annotation export.
2	Backend / Server-Side	Python (FastAPI / Django) Node.js (Optional orchestrator)	• FastAPI / Python: Native ecosystem integration with Google GenAI SDK and image-processing pipelines (Pillow, OpenCV). High asynchronous throughput for streaming LLM outputs. • Handles session workflows, API rate limits, prompt engineering wrappers, and payload optimization.
3	Database / Data Storage	MongoDB / PostgreSQL Firebase / AWS S3 (Blob storage)	• MongoDB / PostgreSQL: Stores user profiles, comic metadata, panel flow structures, JSON prompt histories, and social shares. • Cloud Storage (S3 / Firebase): Handles high-resolution panel image assets and rendered PDF/image comic exports efficiently.
4	Cloud / Hosting / Deployment	Google Cloud Platform (GCP)	• GCP / Cloud Run: Native hosting environment providing low-latency execution and secure IAM access to Google Gemini models.

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		Docker & Kubernetes / Vercel	• Docker: Ensures consistent containerized deployment environments across development, staging, and production.
5	Version Control & CI/CD	GitHub GitHub Actions	• GitHub: Enables seamless team collaboration, code reviews, and branch management. • GitHub Actions: Automates build testing, deployment to GCP/Vercel, and code quality linting on push events.
6	Third-Party APIs / Other Tools	Google Gemini 1.5 Pro / Flash API Imagen 3 / SDXL API WeasyPrint / PDFKit	• Gemini 1.5 Models: Multi-modal narrative generation, dialog scripting, character consistency description maintenance, and panel structuring. • Imagen 3 / Image Gen: Visual rendering of scenes, background environments, and stylized comic book artwork. • WeasyPrint: Assembles finished comic strips and panels into downloadable multi-page PDF comic books.